

Optical Reference Downloader and Lumileds Cage Holder Report

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1 Purpose

This report documents the reusable downloader workflow and the current quick-design reference set for a Lumileds PCB holder that shares 30 mm cage geometry with Hengyang and Thorlabs optical parts.

2 Reusable Downloader

```
python3 cad/tools/optical_reference_downloader.py hengyang
python3 cad/tools/optical_reference_downloader.py hengyang --only gkm-010-polarizer-waveplate-holders
python3 cad/tools/optical_reference_downloader.py thorlabs --preset common-30mm-cage
python3 cad/tools/optical_reference_downloader.py thorlabs SM1L10 CP02T --asset-group Step --asset-group "Cage"
python3 cad/tools/optical_reference_downloader.py report --compile
```

3 Lumileds Holder Geometry

Parameter	Value
body_width_mm	40.0
body_height_mm	41.5
body_thickness_mm	11.0
cage_rod_pitch_mm	30.0
cage_rod_clearance_diameter_mm	6.35
rod_boss_diameter_mm	9.6
light_aperture_diameter_mm	9.6
pcb_outer_diameter_mm	24.0
pcb_pocket_diameter_mm	24.7
pcb_pocket_depth_mm	2.05
pcb_thickness_mm	1.6
pcb_mount_pattern_mm	12.0
pcb_mount_hole_diameter_mm	2.35
bottom_post_mount_hole_diameter_mm	4.2

Reference family: Hengyang Optics 30 mm cage components

Primary reference: GT-090101

Print note: Cage and PCB pockets include practical print clearance; tune after test print.

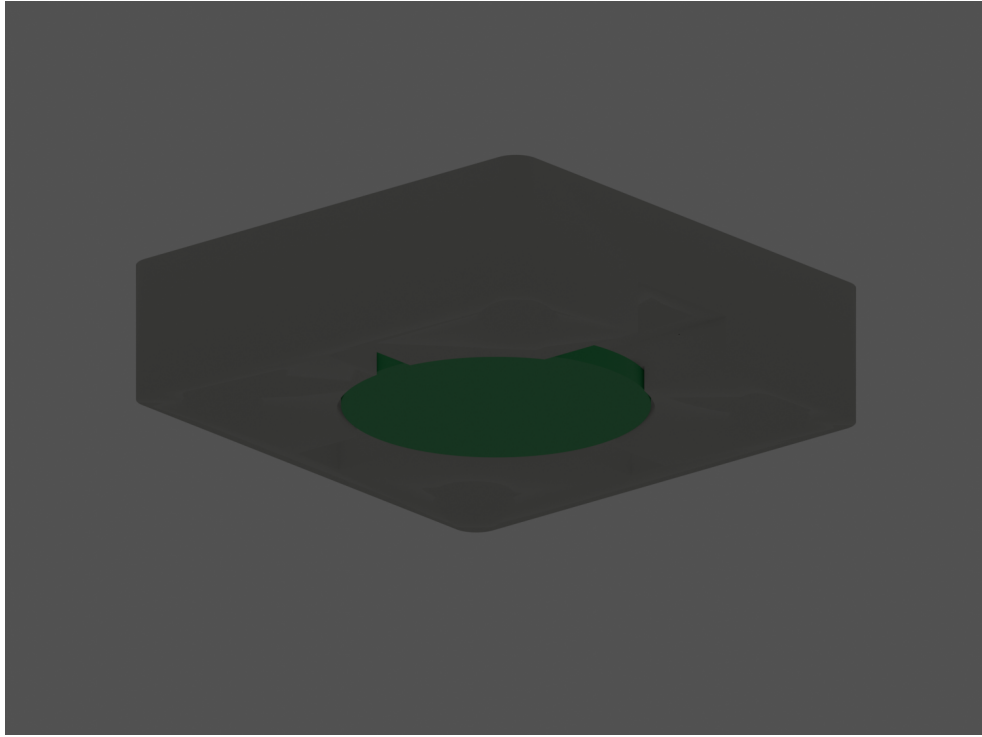
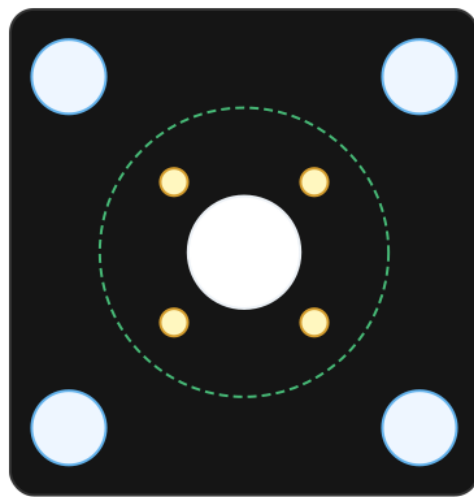


Figure 1: Lumileds holder inspection render.



Independent Lumileds holder

Envelope: 40.0 x 41.5 x 11.0 mm

Cage rods: 30 mm pitch, $\varnothing 6.35$ clearance

PCB pocket: $\varnothing 24.7 \times 2.05$ mm

PCB screws: M2 clearance on 12.0 x 12.0 mm

LED aperture: $\varnothing 9.6$ mm

Reference: GT-090101/HCP/HCT geometry only

Figure 2: Dimension sketch for the holder.

4 Hengyang Reference Set

The Hengyang downloader captures category pages, API JSON, item JSON, drawings, STEP/STP files, and product images. Manifest: `cad/references/hengyang-optics/manifest.json`

Model	Category folder	Local files
GT-090101	gt-090101-waveplate-polarizer	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HCP-22	hcp-22-23-24-lens-cage-plates	drawing-pdf:1, item-json:1, model-step:1, product-image:1
HCP23	hcp-22-23-24-lens-cage-plates	drawing-pdf:1, item-json:1, model-step:1, product-image:1
HCP24	hcp-22-23-24-lens-cage-plates	drawing-pdf:1, item-json:1, model-step:1, product-image:1
HCP-0801	hcp-08-quick-lens-holders	drawing-pdf:1, item-json:1, model-step:1, product-image:1
HCP-0804	hcp-08-quick-lens-holders	drawing-pdf:1, item-json:1, model-step:1, product-image:1
HCP-0805	hcp-08-quick-lens-holders	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
GT-080303	gt-0803-precision-lens-holders	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
GT-080304	gt-0803-precision-lens-holders	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HC-05	hcm-3-beam-splitter-cube	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HC-10	hcm-3-beam-splitter-cube	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HC-12.7	hcm-3-beam-splitter-cube	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HC-20	hcm-3-beam-splitter-cube	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HCM-3	hcm-3-beam-splitter-cube	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HCM-3C	hcm-3-flat-optic-45deg	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HCM-3R	hcm-3-flat-optic-45deg	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HCM-3B3C	hcm-3-flat-optic-45deg	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HCM-3B4C	hcm-3-flat-optic-45deg	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
HKCB1PM	hkcb1pm-right-angle-mirror-holder	drawing-pdf:1, item-json:1, model-stp:1, product-image:1
GKM-0101	gkm-010-polarizer-waveplate-holders	drawing-pdf:1, item-json:1, model-step:1, product-image:1
GKM-0102	gkm-010-polarizer-waveplate-holders	drawing-pdf:1, item-json:1, model-step:1, product-image:1
GKM-0103	gkm-010-polarizer-waveplate-holders	drawing-pdf:1, item-json:1, model-step:1, product-image:1
GKM-0104	gkm-010-polarizer-waveplate-holders	drawing-pdf:1, item-json:1, model-step:1, product-image:1
HCT-150	hct-6mm-cage-rods	drawing-pdf:1, item-json:1, product-image:1
HCT-150-P4	hct-6mm-cage-rods	drawing-pdf:1, item-json:1, product-image:1

HCT-150-P8	hct-6mm-cage-rods	drawing-pdf:1, item-json:1, product-image:1
HCT-010	hct-6mm-cage-rods	drawing-pdf:1, item-json:1, model-step:1, product-image:1
HCT-010-P4	hct-6mm-cage-rods	drawing-pdf:1, item-json:1, model-step:1, product-image:1
HCT-010-P8	hct-6mm-cage-rods	drawing-pdf:1, item-json:1, model-step:1, product-image:1

5 Thorlabs Reference Set

The Thorlabs downloader queries public product metadata and stores selected CAD assets by part number. Manifest: `cad/references/thorlabs-optics/manifest.json`

Part	Name	Local files
C4W	C4W	CAD DXF:1, CAD PDF:1, Step:1, product-image:1, product-json:1
CP02T	CP02T	CAD DXF:1, CAD PDF:1, Step:1, product-image:1, product-json:1
ER1	ER1	CAD DXF:1, CAD PDF:1, Step:1, product-image:1, product-json:1
ER2	ER2	CAD DXF:1, CAD PDF:1, Step:1, product-image:1, product-json:1
KCB1	KCB1	CAD DXF:1, CAD PDF:1, Step:1, product-image:1, product-json:1
KM100	KM100	CAD DXF:1, CAD PDF:1, Step:1, product-image:1, product-json:1
LCP02	LCP02	CAD DXF:1, CAD PDF:1, Step:1, product-image:1, product-json:1
SM1L10	SM1L10	CAD DXF:1, CAD PDF:1, Step:1, product-image:1, product-json:1

6 Measured STEP Bounds

STEP bounds are approximate imported bounding boxes, useful for fast layout planning before detailed inspection.

STEP file	X mm	Y mm	Z mm
cad/references/hengyang-optics/gkm-010-polarizer-waveplate-holders/GKM-0101/GKM-0101_model-step.step	30.0	38.5	12.5
cad/references/hengyang-optics/gkm-010-polarizer-waveplate-holders/GKM-0102/GKM-0102_model-step.step	42.0	51.0	12.5
cad/references/hengyang-optics/gkm-010-polarizer-waveplate-holders/GKM-0103/GKM-0103_model-step.step	48.0	58.5	12.5
cad/references/hengyang-optics/gkm-010-polarizer-waveplate-holders/GKM-0104/GKM-0104_model-step.step	70.0	78.761	16.0
cad/references/hengyang-optics/gt-0803-precision-lens-holders/GT-080303/GT-080303_model-stp.stp	59.944	59.944	35.738
cad/references/hengyang-optics/gt-0803-precision-lens-holders/GT-080304/GT-080304_model-stp.stp	59.944	59.944	38.151
cad/references/hengyang-optics/gt-090101-waveplate-polarizer/GT-090101/GT-090101_model-stp.stp	40.0	41.5	18.0
cad/references/hengyang-optics/hcm-3-beam-splitter-cube/HC-05/HC-05_model-stp.stp	25.4	25.4	25.4

cad/references/hengyang-optics/hcm-3-beam-splitter-cube/HC-10/HC-10_model-stp.stp	25.4	25.4	25.4
cad/references/hengyang-optics/hcm-3-beam-splitter-cube/HC-12.7/HC-12.7_model-stp.stp	25.4	25.4	25.4
cad/references/thorlabs-optics/C4W/C4W_Step_C4W-E0W.step	50.8	39.624	50.8
cad/references/thorlabs-optics/CP02T/CP02T_Step_1703-E0W.step	41.402	40.64	12.7
cad/references/thorlabs-optics/ER1/ER1_Step_0775-E0W.step	5.994	30.48	5.994
cad/references/thorlabs-optics/ER2/ER2_Step_0780-E0W.step	5.994	55.372	5.994
cad/references/thorlabs-optics/KCB1/KCB1_Step_13038-E0W.step	51.13	48.26	51.103
cad/references/thorlabs-optics/KM100/KM100_Step_6628-E0W.step	52.451	39.311	52.451
cad/references/thorlabs-optics/LCP02/LCP02_Step_2869-E0W.step	70.612	70.612	12.7
cad/references/thorlabs-optics/SM1L10/SM1L10_Step_0371-E0W.step	29.21	30.48	30.48

7 Agent Contract

1. Search or identify vendor part numbers, then download vendor material into `cad/references/`.
2. Keep raw pages, JSON, drawings, STEP files, images, and manifests together.
3. Measure STEP bounds when a CAD kernel is available, but verify critical dimensions in vendor drawings.
4. Use downloaded references only as design constraints. New printable parts should remain independent designs.
5. Regenerate this PDF after adding reference families or changing the Lumileds holder.

8 Key Local Outputs

holder STEP: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder.step`
holder STL: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder.stl`
assembly STEP: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder.assembly`
assembly STL: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder.assembly.stl`
dimension sketch SVG: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder_dimension_sketch.svg`
dimension sketch PNG: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder_dimension_sketch.png`
dimension sketch PDF: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder_dimension_sketch.pdf`
inspection render PNG: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder_inspection_render.png`
Blender inspection scene: `cad/designs/lumileds_hengyang_30mm_cage_holder/artifacts/lumileds_hengyang_30mm_cage_holder_inspection_scene.blend`